David Alm

PhD Student in Machine Learning for Molecular Biology
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EDUCATION

Aug. 2026 - (June 2030) **PhD in Machine Learning, KTH Royal Institute of Technology**
School of Electrical Engineering and Computer Science (EECS)
Department of Robotics, Perception, and Learning (RPL).

Supervisors: Hossein Azizpour (Main @ RPL), Joakim Andén (Co @ Math Department), Marta Carroni (Co @ Stockholm University (SU)).

I work on generative foundational models over flexible 3D macromolecular structure as priors in downstream heterogeneous cryoelectron-micropscopy (cryo-EM) reconstructions. My work is part of Digital Future's flagship project on recovering the dynamics of biological macromolecules using cryo-EM data with data-driven priors, connecting Marta and her group's domain expertise and data curation capabilities at SU and SciLifeLab, Hossein and his group's expertise in data-driven generative modeling in life science applications, and Joakim's knowledge in statistical inverse problems and 3D reconstruction methods in cryo-EM.

Aug. 2025 - Feb. 2026 **Exchange Semester, ETH Zürich**
Department of Mathematics. GPA: 5.6 / 6.0. Courses: AI in the Sciences and Engineering by Prof. S. Mishra (6 ECTS), Optimization for Data Science by Prof. N. He (10 ECTS), Probabilistic Artificial Intelligence by Prof. A. Krause (8 ECTS).

Aug. 2024 - June 2026 **MSc within Applied and Computational Mathematics, KTH Royal Institute of Technology**
Track: The Mathematics of Data Science. GPA: 4.95 / 5.0.

Aug. 2021 - June 2024 **BSc within Engineering Mathematics, KTH Royal Institute of Technology**
Thesis within sports betting and artificial neural networks: DiVa. GPA: 4.8 / 5.0.

Aug. 2022 - June 2023 **BSc within Business and Economics, SSE Stockholm School of Economics**
Award for Outstanding Performance in BE501 Microeconomics. Dropped out. GPA: 4.4 / 5.0.

Aug. 2018 - June 2021 **Upper Secondary School, Natural Sciences and Entrepreneurship, Östra Reals Gymnasium**
Grades: 19.88 / 20.00 + 2.50 / 2.50.

Aug. 2008 - June 2018 **Elementary School, IB Bilingual Program, Vasa International School of Stockholm**
Grades: 340 / 340

WORK EXPERIENCE

June 2025 - Aug. 2025 **ML Research Intern, Aalto University**
Part of Professor Dominik Baumann's Cyber-Physical Systems group at the Department of Electrical Engineering and Automation. Together with Prof. Baumann's PhD student Mingwei (Mark) Deng, I explored ways of learning the causal mask in a modular MLP architecture to learn the causal dynamics of MDPs using interventions, with the aim of improving meta-transfer in model-based RL.
Referee: Prof. Dominik Baumann, dominik.baumann@aalto.fi.

Aug. 2024 - Jan. 2025 **Teaching Assistant, KTH Royal Institute of Technology**
I held exercise and lab sessions, helped with grading, among other things within the following classes; SF1930 Statistical Learning and Data Analysis and SF1918 Probability Theory and Statistics.
Referee: Prof. Liam Solus, solus@kth.se. Prof. Joakim Andén, janden@kth.se.

Sep. 2023 - Sep. 2024 **Trading Supervisor and Quantitative Analyst, Ampfield AB**
Ampfield is an algorithmic proprietary trading firm that trades futures in FX, agriculture and other commodities. These contracts are traded almost 24 hours a day and sometimes Ampfield's trading system encounters problems. My role as a trading supervisor is to supervise the system and respond when necessary. During the summer of 2025, I had a project as a full time quantitative analyst to investigate one of the firm's hypotheses within equities.
Referee: COO Henrik Melander, henrik.melander@ampfield.se.

Sep. 2021 - Apr. 2022 **Project Manager of Fusion 2022, The Physics Chapter at KTH**
Fusion is the Physics Chapter at KTH's annual career fair. I managed a project group consisting of 13 students and oversaw everything from marketing and sales to financials and logistics.

June 2020 - June 2021 **CEO and Co-Founder of Hemply UF**
Company within the framework of UF, Ung Företagsamhet, the Swedish counterpart to JAW, Junior Achievement Worldwide. We generated ca. 400 000 SEK in revenue during 3 months of sales and received several regional and national awards.

AWARDS, HONORS & SCHOLARSHIPS

2026 Scholarship from the general student scholarship for outstanding grades at KTH

2025 Scholarship from the Henrik Göransson Sandviken Foundation Scholarship fund for outstanding grades at KTH

2023 Outstanding Performance in BE501 Microenomics at SSE

2021 Several regional and national awards for Hemply UF

TECHNICAL SKILLS

Theory	Deep Learning; Probabilistic Machine Learning; Generative Modeling; Probability Theory; Optimization; Bayesian Inverse Problems
Techniques	Object-Oriented Programming; Functional Programming; Unit Testing; CI/CD; High-Performance Computing
Programming Languages	Python; C++; Java; SQL; MatLab; LaTeX
Libraries	NumPy; JAX; SciKit-Learn; SciPy; Pandas; Matplotlib; PyTorch; Weights & Biases
Tools	Git; Slurm; Bash; SSH